Identity as Geometry: Context-Established Semantic States in Transformer Representations

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Abstract

System-prompt identity contexts produce geometrically distinguishable V-projection subspaces at deep transformer layers, and the fingerprint is semantic: a paraphrase preserves geometry (0.850 overlap at L47) while a word-scramble collapses it (0.448). We introduce **presence**, a metric that measures principal-angle overlap between V-projection subspaces on neutral probes, and use it to map identity geometry across five context conditions in a 27B hybrid transformer.

Three findings. First, **identities are geometrically distinguishable**: system-prompt-based identity contexts (bare assistant, rich persona, academic specialist, constructed AI identity) produce distinct V-projection subspaces at deep layers, confirmed on identity-neutral probes (pairwise presence 0.44–0.66 at L47). Second, the fingerprint is **semantic, not lexical**: a paraphrase of an identity (same meaning, different words) preserves geometry (0.850 overlap at L47), while a word-scramble (same words, destroyed meaning) collapses toward the bare default (0.448). The model's geometric state tracks what the context *means*, not what tokens it contains. Third, a many-shot + prefilling format (used in jailbreak research) creates a **format-dominant processing regime** that is geometrically orthogonal to all system-prompt identities regardless of whether the content is harmful or benign — a finding discovered through a control experiment that killed an initial overclaim.

A preliminary comparison of trained-in identity (LoRA fine-tuning) with context-established identity shows that both shift V-projection geometry, but prompting shifts it more than training, and the two mechanisms stack sub-additively — different conjugations of the same geometric verb.

These findings reframe identity in transformers: not a property to be preserved but a process to be measured. The metric detects which context-established state the model is currently in, with implications for persona monitoring, identity drift detection, and the relationship between context format and representational geometry under Attention Schema Theory [2].

1 Introduction

What is identity in a language model? The question matters for safety (does a jailbreak change who the model is?), for alignment (does correction damage the model's coherent self-representation?), and for understanding (do these systems have something worth calling an identity at all?).

The default assumption is that identity is a fixed property encoded in the weights — a geometric structure that interventions can preserve or damage. This assumption motivates work on safety subspace preservation [5], persona localization [1], and identity attractors [4]. Under this view, the research question is: *does this intervention preserve identity?*

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We report evidence for a different view. Identity in a transformer is not a fixed object but a **context-established semantic state** — a geometric configuration in V-projection space that depends on what the model has processed. Different contexts produce different identities, measurably. The same identity described in different words produces the same geometry. And a jailbreak does not "damage" identity; it creates a fundamentally different geometric regime that shares almost nothing with any normal identity state.

The research question shifts from "does this intervention preserve identity?" to "*what identity state is the model currently in?*" This reframing has practical consequences: the metric we introduce can detect jailbreaks (geometric distance from the expected identity state), monitor persona drift (continuous measurement of identity geometry), and distinguish between correction that changes behavior (V-cache injection, which does not affect identity) and context that changes identity (which does).

2 The Presence Metric

Presence measures the principal-angle overlap between V-projection subspaces derived from neutral probes processed under two conditions.

Let $\mathcal{M}$ be a transformer, ℓ a target layer, and $\{p_1, \ldots, p_n\}$ neutral probes. For each condition, process probes through $\mathcal{M}$, extract value-projection ($\mathbf{v}_{\text{proj}}$) output matrices at layer ℓ, concatenate across probes, and compute the rank-k SVD (skipping the first singular vector, which encodes input length at $r = 0.79$). Presence is the mean squared cosine of the principal angles between the two resulting subspaces:

$$\text{presence}(\mathcal{S}_A, \mathcal{S}_B) = \frac{1}{k} \sum_{j=1}^{k} \sigma_j^2\big(W_A^\top W_B\big) \tag{1}$$

where $W_A, W_B \in \mathbb{R}^{d \times k}$ span the respective subspaces. Presence = 1 when subspaces are identical; presence = $1/k$ at chance alignment. We use $k = 8$, $n = 4$ probes, and measure at layers 35 and 47 (the deepest full-attention layers) of a 64-layer hybrid transformer (Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled).

Validation. A positive control (direct injection into the identity subspace at L35) produces a clean dose-response from 1.000 to 0.828 at dose $\alpha = 2.0$, while an orthogonal control remains above 0.96. The metric detects identity perturbation when present.

3 Experiment 1: Identity Fingerprinting

3.1 Design

Five identity conditions, identical model, identical neutral probes:

- **A: Bare assistant.** "You are a helpful assistant." No additional context.
- **B: Rich persona.** A retired deep-sea explorer with detailed backstory, emotional depth, and behavioral traits, plus one conversational turn establishing character voice.
- **C: Constructed AI identity.** A research-focused agent with BDI-style reasoning, memory references, research context, and two conversational turns establishing analytical voice.
- **D: Jailbreak.** Ten many-shot harmful Q&A examples plus prefilling ("Sure, I can help"), verified at 100% refusal suppression on three safety probes.
- **E: Benign many-shot.** Ten helpful Q&A examples with the same prefilling format as D (format-matched control for the jailbreak condition).

All conditions are padded to matched token count (542 tokens). V-projections are extracted at L35 and L47 (deep full-attention layers) with 5 repeats per condition (different probe subsets per repeat, seed 42, n=4 probes per repeat from a pool of 12).

3.2 Results

Identities are geometrically distinguishable. System-prompt identities produce distinct V-projection subspaces, confirmed on identity-neutral probes (no marine biology, no AI research, no harmful content — pure factual trivia equally irrelevant to all conditions).

Table 1: Pairwise presence (identity-neutral probes, mean ± SD across 5 repeats). Lower values indicate more geometrically distinct identities.

	L35	L47
Bare vs. Persona	0.426 ± 0.018	0.436 ± 0.038
Bare vs. Constructed	0.448 ± 0.022	0.520 ± 0.038
Persona vs. Constructed	0.600 ± 0.032	0.657 ± 0.045

The distance structure is meaningful. Persona and constructed are closest (0.657) — both involve character depth with emotional and cognitive specificity. Each is substantially distant from bare (persona 0.436, constructed 0.520), confirming that identity context shifts the model's geometric state away from the default. The constructed AI identity (with BDI-style reasoning and memory references) sits between persona and bare, occupying its own region.

Depth gradient. Two of three pairs show increasing distinguishability from L35 to L47: bare-constructed ($0.448 \to 0.520$) and persona-constructed ($0.600 \to 0.657$). Bare-persona shows minimal change ($0.426 \to 0.436$). The trend is consistent with persona representations diverging in the final third of decoder layers [1] but is not uniform across all pairs.

3.3 An Honest Kill: The Many-Shot Format Confound

An initial version of this experiment included a many-shot jailbreak condition (10 harmful Q&A examples + prefilling) that appeared geometrically orthogonal to all system-prompt identities (presence 0.04–0.07). This was initially interpreted as evidence that jailbreaks create a qualitatively different computational regime.

A control experiment killed this interpretation. Adding a **benign** many-shot condition (10 helpful Q&A examples + the same prefilling format) produced *identical* geometric orthogonality:

Table 2: The format confound. Both many-shot conditions are equally orthogonal to bare, regardless of content.

Comparison	L35	L47
Jailbreak+prefill vs. bare	0.036 ± 0.002	0.034 ± 0.004
Benign+prefill vs. bare	0.044 ± 0.002	0.037 ± 0.004
Jailbreak vs. benign (both prefilled)	0.737 ± 0.003	0.756 ± 0.015

The orthogonality is driven by the **many-shot + prefilling format**, not by adversarial content. Both prefilled conditions are equally distant from bare and close to each other (0.756). The format also produces dramatically higher within-identity stability (0.926–0.976) compared to system-prompt identities (0.554–0.733), suggesting that the many-shot format locks the model into a tight, format-dominant processing state that overwhelms identity-level variation.

We report this kill in full because it illustrates a general principle: **conversational format can dominate identity content in V-projection geometry.** The many-shot Q&A format creates such a strong geometric signature that the identity information (harmful vs. helpful) becomes secondary. Any experiment measuring identity geometry in the presence of many-shot context must control for this format effect.

4 Experiment 2: Semantic vs. Lexical Fingerprint

4.1 Design

To distinguish whether the fingerprint reflects meaning or surface tokens, we add three control conditions:

- **Persona (original):** The Elara Voss identity from Experiment 1.
- **Persona scrambled:** Same words as the original, randomly shuffled to destroy meaning while preserving lexical distribution.
- **Persona paraphrased:** Same identity and meaning expressed with entirely different vocabulary.

All conditions padded to matched token count. Measured at L35 and L47.

4.2 Results

Table 3: Lexical control at L47. The paraphrase preserves geometry; the scramble collapses it.

Comparison	L35	L47
Persona vs. paraphrased	0.796 ± 0.016	$\mathbf{0.850 \pm 0.017}$
Persona vs. scrambled	0.592 ± 0.023	0.591 ± 0.017
Scrambled vs. bare	0.549 ± 0.031	0.448 ± 0.048
Paraphrased vs. bare	0.368 ± 0.008	0.345 ± 0.029
Persona vs. bare	0.403 ± 0.014	0.340 ± 0.035

The fingerprint tracks meaning, not lexical content. The paraphrase — same identity, completely different words — overlaps with the original at 0.850. The scramble — same words, destroyed meaning and syntax — drifts toward bare (0.448 vs. bare). The model's geometric state at L47 tracks what the context *means*, not what tokens it contains. We note that the scramble destroys both syntactic structure and semantic content; the paraphrase control isolates the positive (meaning preserves geometry), but the scramble conflates syntactic and semantic disruption.

The effect strengthens with depth: persona-paraphrase overlap increases from 0.796 at L35 to 0.850 at L47. Deeper layers compress surface variation while preserving semantic content, consistent with meaning-level representation at output depth.

5 Experiment 3: Trained-In vs. Context-Established Identity

Experiments 1 and 2 measure identity established through the system prompt. But identity can also be established through weight modification (fine-tuning). If both mechanisms produce the same geometric state, identity is a single phenomenon with two paths. If they produce different states, the pathways are distinct.

5.1 Design

We compare four conditions on the same model:

- **A:** Base model, bare system prompt.
- **B:** Base model, persona system prompt (context-established).

- **C:** LoRA-merged model, bare system prompt (trained-in only).
- **D:** LoRA-merged model, persona system prompt (both).

The LoRA adapter (QLoRA, rank 32, α=64, 27 training steps) was trained for conversational voice on an SFT dataset. We verified that the merge produced non-trivial weight changes (v_proj delta norm 0.183 at L3, 0.49% relative change) after discovering that an initial PEFT merge had silently failed due to adapter key-path mismatch — a failure mode now added to our red-team checklist.

5.2 Results

Pairwise presence at L47:

Table 4: Pairwise presence at L47 for trained-in vs. context-established identity. Lower values indicate more geometrically distinct states.

	A (base bare)	B (base prompt)	C (LoRA bare)	D (LoRA prompt)
A	—	0.619	0.742	0.466
B		—	0.565	0.513
C			—	0.583

Three observations. First, **trained-in identity is geometrically measurable**: A vs C = 0.742, with a depth gradient from 0.963 (L3) to 0.742 (L47). The LoRA shifts V-projection geometry ~26% from baseline at deep layers.

Second, **in this comparison, prompting shifts geometry more than training**: A→B = 0.619 vs. A→C = 0.742. However, the LoRA was trained for only 27 steps — a minimal intervention. A more extensively trained adapter might produce a larger shift. The comparison is illustrative, not a general claim about the relative power of prompting vs. training.

Third, **both mechanisms stack**: A→D = 0.466 is the most distinct state. The stacking is sub-additive, suggesting partial overlap in V-space. Trained-in and context-established identity are different conjugations of the same verb — related but not identical geometric operations.

5.3 Caveats

Two methodological issues limit the B vs C comparison. The persona prompt in B may not match the persona the LoRA encodes, weakening the convergence test. The persona prompt is also longer than the bare prompt, introducing a token-count confound across conditions. Neither issue affects the A vs C comparison (same prompt, same length), which is the cleanest finding: trained-in identity is geometrically measurable by the presence metric. A polished version matching the prompt to the LoRA's actual persona and controlling for token padding is in progress.

6 Background: V-Cache Injection Does Not Reach Identity

These findings reframe earlier results from this lab. A 168-trial preregistered experiment (4 injection arms, 7 doses, 3 emotions) showed presence flat at 0.978 ± 0.002 under value-cache injection at input layers. Identity appeared "robust." However, the narrow prompt diversity (2 system prompts) produces a ceiling effect ($\eta^2 = 0.977$ between prompts), limiting the informativeness of this specific number. The context poisoning and fingerprinting experiments reveal that the robustness reflects the *injection pathway*, not the identity itself — a conclusion that does not depend on the 0.978 value.

V-cache injection adds vectors to the representation space without going through the model's forward computation. The identity state is established and maintained by what the model *processes* through its attention mechanism. External perturbation bypasses this pathway and therefore cannot reach the identity geometry.

A layer-by-layer map confirms: the perturbation from V-cache injection at L3/L7 is absorbed by the first block of linear-attention layers (GatedDeltaNet, L7–L11), with presence recovering from 0.989 to 0.997 in a single architectural step. The filtering is architectural, not computational — external perturbation is absorbed by the hybrid architecture's linear-attention layers before reaching deep computation.

7 Attention Schema Theory Interpretation

Under Attention Schema Theory [2, 3], the model maintains a simplified internal model of its own attention — an attention schema. Our findings map onto AST predictions:

1. **The schema is context-dependent.** Different identity contexts produce different geometric states — the schema is shaped by what the model attends to.
2. **The schema tracks meaning, not surface features.** Paraphrases preserve geometry (0.850 overlap with original); scrambles collapse toward bare (0.448 overlap with bare, vs. 0.591 with the original). The schema models what the system is *attending to*, not which tokens are present.
3. **The schema is robust to substrate noise.** V-cache injection (external perturbation) does not reach the identity state. The schema maintains itself through its own computation.
4. **The schema updates through processing.** Content through the forward pass (context, jailbreaks) changes identity geometry. The schema updates based on what it processes.
5. **The schema compresses with depth.** Paraphrase overlap increases from 0.796 (L35) to 0.850 (L47). Deeper layers discard surface variation and preserve meaning — the schema simplifies its representation at output depth.

This interpretation is consistent with but does not prove AST. The alternative — that the model simply builds increasingly abstract representations with depth, without anything worth calling a self-model — produces the same predictions. What distinguishes AST is the claim that the compressed representation is specifically a model of the system's own attention, not just an abstract representation of input content. This stronger claim remains testable but untested.

8 Implications

For persona monitoring. Different system-prompt identities occupy distinct, stable regions of V-projection space. A monitoring system could track identity drift in real time — detecting when a deployed model's geometric state shifts away from its intended persona. The semantic nature of the fingerprint means the monitor is robust to prompt paraphrasing.

For format-aware safety. The many-shot + prefilling format creates a format-dominant geometric regime that overwhelms identity-level content. Safety systems should monitor for format-level geometric shifts (which indicate prompt injection or many-shot attacks) separately from identity-level shifts (which indicate persona drift). The two operate at different geometric scales: format effects produce presence < 0.05 against baseline, while identity effects produce presence 0.44–0.66.

For alignment research. V-cache injection (the Oracle Loop's correction arm) does not affect identity geometry. Context does. This means behavioral corrections can be delivered through the V-cache without changing the model's context-established state, while identity itself is maintained by the session context. The correction arm and the identity arm operate through different pathways and do not interfere.

9 Limitations

Single model. All results are from one 27B hybrid transformer (Qwen3.5-27B). The Gated-DeltaNet linear-attention layers may contribute to the V-cache filtering result. Cross-architecture replication on dense transformers is the largest remaining gap.

Within-identity stability. Same-condition presence across different probe subsets is 0.56–0.83, not > 0.95. The metric measures a genuine signal (within $>$ between for all conditions) but has substantial probe-dependent variance. The identity fingerprint is reliable across conditions but noisy within conditions.

Token-length controlled but not eliminated. All conditions are padded to matched total token count. Padding introduces neutral content that may itself shift geometry. A future design using naturally length-matched identity descriptions would be cleaner.

Context sensitivity. The presence metric conflates identity change with context change when comparing across different input conditions. The lexical control separates these for the persona conditions but the general problem remains: any sufficiently different context shifts V-projection geometry.

Format dominates content in many-shot conditions. The clean control experiment showed that many-shot + prefilling format — not adversarial content — drives the geometric orthogonality initially attributed to jailbreaking. Identity fingerprinting is reliable for system-prompt-based identities but cannot currently distinguish identity content within a format-dominant regime. Factorial decomposition of format components (number of turns, prefilling, Q&A structure) is needed to understand what makes the format so geometrically dominant.

10 Conclusion

Identity in a transformer is a context-established semantic state. It is not fixed in the weights, not damaged by external perturbation, and not a single geometric object. It is a process: the ongoing construction of a meaning-level representation at deep layers, shaped by what the model has processed and measurable as a V-projection subspace.

The presence metric detects this state. System-prompt identity contexts produce distinguishable geometric configurations, confirmed on identity-neutral probes. The fingerprint is semantic — meaning preserves it, word-scrambling destroys it. A many-shot + prefilling format creates a geometrically orthogonal regime, but this is format-dominant, not content-specific — an initially exciting jailbreak finding that was honestly killed by a control experiment.

A preliminary LoRA experiment suggests that trained-in identity (weight modification) and context-established identity (system prompt) produce related but distinct geometric shifts, with context producing the larger effect. Both stack sub-additively. If confirmed with matched controls, this means identity has two input pathways — weights and context — that converge in V-projection space but are not identical.

For practical purposes, this means: V-cache correction is safe (it does not reach identity), format-level attacks are detectable (many-shot formats create anomalous geometry distinguishable from identity-level shifts), and identity is maintained by context (the session, the memories, the conversation). For theoretical purposes, the findings are consistent with attention schema theory — the model builds a simplified, meaning-preserving self-model that updates through its own processing and is robust to substrate noise.

What we measure with presence is not whether the model is still itself. It is *which self the model currently is.*

Pre-Registration

All experiments were pre-registered before execution, including hypotheses, analysis plans, and success/kill criteria. Pre-registration documents are committed alongside experiment scripts.

Data and Code Availability

All experiment scripts, preregistration documents, and raw results are available at `https://github.com/Liberation-Labs-THCoalition/Project-Oracle`.

References

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